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Microhardness, indentation toughness, elasticity, plasticity, and brittleness of Ge–Sb–Se chalcogenide glasses

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Abstract

Elastic/plastic properties of Ge–Se binary and Ge–Sb–Se ternary chalcogenide glasses were explored by indenting with Vickers and Brinell (spherical) microindenters using static and 'instrumented' (load versus penetration) recording machines and analyzed further using a non-contact profilometer. The load-dependence of microhardness, indentation fracture toughness and brittleness versus the average covalent coordination number, $\langle r \rangle$, were studied. It is concluded that the elastic recovery of deformations is maximized at $\sim GeSe_4$ ($\langle r \rangle = 2.4$), which is consistent with 'optimized connectivity' of the rigidity percolation arguments. Surprisingly, many of the extrema were nearly non-existent in the ternary glasses.

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Keywords: Chalcogenides; Fracture; Hardness; Indentation; Microindentation; Optical microscopy

1. Introduction

The atom-averaged covalent coordination number, $\langle r \rangle$, of chalcogenide glasses is given by the Mott's '$8 - N$ rule', where N is the number of outer shell electron in a given component element, and is used to describe the crosslinking between the atoms. Based on topological constraints counting methods, Phillips [1] proposed that optimum network connectivity exists at $\langle r \rangle = 2.4$, where the number of degrees of freedom just exhausts the number of constraints imposed by bond-bending and bond-stretching forces. Thorpe [2] suggested that, below $\langle r \rangle = 2.4$, the glass network is composed of floppy or zero-frequency vibrational modes and is unbraced or underconstrained. The converse is true above 2.4, where the glass is overbraced or overconstrained. At $\langle r \rangle = 2.4$, the number of zero-frequency modes should diminish to zero resulting in rigidity to percolate. Later, Tanaka [3] showed that, at $\langle r \rangle = 2.67$, there exists yet another structural phase transition. Covalent glasses having $\langle r \rangle < 2.67$ may have low-dimensional structures, whereas $\langle r \rangle > 2.67$ glasses would be cross-linked 3-dimensionally. Thus, there are two separate topological thresholds that may be observed in chalcogenide glasses when their properties are studied as a function of the coordination number: one is at $\langle r \rangle = 2.4$, the 'rigidity percolation' threshold, and the other at $\langle r \rangle = 2.67$, the '2D $\rightarrow$ 3D' threshold. In binary Ge–Se glasses, the $\langle r \rangle = 2.67$ coincides with the stoichiometric compound $GeSe_2$ with complete heteropolar bonding. Hence, the $\langle r \rangle = 2.67$ threshold in Ge–Se system is also a 'chemical threshold'.

Measurement of the properties of chalcogenide glasses as a function of $\langle r \rangle$ has been the subject of many studies. A summary of the results on several physical properties such as the elastic moduli, molar volume, Poisson's ratio, Vickers hardness, indentation toughness and glass transition temperature can be found in Refs. [4–6]. Of the properties mentioned above, only molar volume shows a distinct minimum at $\langle r \rangle = 2.4$. However, the study of second derivative thermodynamic properties such as the isothermal compressibility, thermal expansion coefficient,

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and heat capacity during glass transition as well as in the supercooled liquid state leads to the conclusion that $\langle r \rangle = 2.4$ glasses undergo minimum configurational changes or the least structural relaxation to form an optimum glass [7,8]. Recent publications by Boolchand's group [9,10] suggest that $\langle r \rangle = 2.4$ glasses are thermally reversible ($\Delta H_{nr} \sim 0$) due to structural self-organization of disordered networks. In recent years, it has been recognized [11,12] that the indentation hardness and the associated indentation fracture toughness as measured using an instrumented (load versus penetration) recording microindenter show extrema when the contribution of network plasticity during deformation is subtracted out.

When an indenter penetrates a solid, the applied load produces a stress field which can be considered to be composed of two components: a *hydrostatic* component which leads to densification without a change in shape and a *deviatoric* component (pure shear) which is volume-conserving but changes the shape. Each of these components of the stress field first causes an elastic deformation. After exceeding a yield strength, each of the components continues to deform the material plastically. Note that the yield strength in pure hydrostatic loading need not equal the yield strength in pure shear. Upon unloading, the elastic portion of total deformations may recover; the plastic portion is not recovered. (Some of the elastic deformation may not recover because of the enforcement of the compatibility criteria.) An essential feature of elastic and plastic deformations is the time-independence. When a material is viscoelastic, deformations become time-dependent and can be simulated by mechanical analogs comprising Hookean springs and Newtonian dashpots [13]. Upon unloading, the viscous deformation of a dashpot is recoverable only if it is associated with an elastic spring in parallel (Voigt–Kelvin element), which is termed 'delayed elasticity'. Creep of an unconstrained dashpot is unrecoverable.

Traditional hardness property is measured using a static machine and involves the measurement of the diagonal of the impression after unloading. The Vickers hardness H_v is obtained as the vertically applied load P divided by contact (slant) area of the square diamond pyramid impression:

$$H_v = 2P\sin(\theta/2)/d^2, \tag{1}$$

where d is the average diagonal of the impression and θ is the included angle between the faces of the square pyramid ($\theta = 136°$ for a Vickers indenter). Hardness that is calculated as load divided by the curved area of the impression after indenting a smooth surface with a rigid sphere is called Brinell hardness H_B and is given by

$$H_B = 2P/\{\pi D[D - \sqrt{(D^2 - d^2)}]\}, \tag{2}$$

where D is the indenter diameter and d is the diameter of the impression (projection) circle. Note that, in each case, the computation of the slant area or the curved area assumes that the impression has the same geometry as the indenter.

Continuous load versus penetration sensing ('recording' or 'instrumented') microindenters enable the generation of stress–strain curves during both loading and unloading. For a recording microindenter, different Brinell hardness values such as Brinell hardness-under-load (H_{BL}) from the maximum penetration depth h_{max} and Brinell hardness-from-unloaded-depth (H_{BD}) from the unloaded penetration depth h_o can be calculated by

$$H_{BL} = \frac{P_{max}}{\pi D h_{max}}, \tag{3}$$

$$H_{BD} = P_{max}\pi D h_o, \tag{4}$$

where P_{max} is the maximum load. The denominator in each case is the estimate of the curved area of the indentation. Clearly, there are issues as to what the true contact area ought to be, particularly in view of a pile-up ('mound') or a sinking-in that may occur during the indentation process. The work of Chaudhri and Winter [14] suggests that the load is supported by the sides of the pile-up region (or less the sink-in region), hence, the contact area should be estimated taking any pile-up region into account.

The indentation fracture toughness K_c is calculated using the equation of Anstis et al. [15]

$$K_c = \Psi(E/H)^{1/2}(P/c_o^{3/2}), \tag{5}$$

where Ψ is a material-independent constant (=0.016 for Vickers produced radial cracks), c_o is the average length of the radial crack. Lawn and Marshall [16] defined brittleness B as the ratio of hardness to fracture toughness (=H/K), from which Sehgal and Ito [17] developed the expression,

$$B = [(\alpha_o)^{-3/4}\beta_o H^{1/4}](c_o/a)^{3/2}P^{-1/4} \tag{6}$$

where α_o and β_o are geometrical parameters. Using SI values of hardness, the term in the square brackets can be approximated by 2386 $N^{1/4}\,m^{-1/2}$.

In this study, the earlier research [11,12] on binary Ge–Se and ternary Ge–Sb–Se chalcogenide glasses using static and recording instruments and a Vickers indenter was extended to using 400 μm, 1 mm and 2 mm radius Brinell indenters. Microhardness, indentation toughness and brittleness were obtained. A non-contact laser profilometer was used to scan the landscape profile across the indentation and to obtain a topological view of the indent and fractures emanating from the indent. The profilometer data are analyzed for elastic and plastic/flow deformations as a function of the average covalent coordination number of the glass.

2. Experimental procedure

The procedures for fabricating chalcogenide glasses from raw materials are given elsewhere [18]. Binary Ge–Se and ternary Ge–Sb–Se glass samples, which were in good condition, were selected out of a stock of glasses melted by two prior investigators [19,20] and used for this study. These covered a range of average coordination numbers $\langle r \rangle$ from 2.0 to 2.8. Samples were ground flat to form disks using 600 and 1000 SiC grits and polished with 0.3 μm alumina powder to obtain surfaces parallel to within 2°. The diameter and thickness of the samples ranged 8–15 mm and 2–6 mm respectively.

A Buehler Micromet II hardness tester fitted with a Vickers' diamond indenter was used for making static indentation measurements. Each sample was indented with loads of 50, 100, 200, 300, and 500 g. The loading time was 15 s in order to include the observations of possible viscous flow. The resulting impressions and crack diagonals were measured using a microscope attached to a video camera and an image analysis software. Indentations were also performed at −80 °C by immersing specimens in a liquid N_2-cooled methanol bath.

Glass specimens were then indented using the load and depth sensing dynamic microindenter. Each dynamic indentation was carried out to loads of 1 kg or 1.5 kg. The indenter was driven at a rate of 0.2 μm/s with 15 s dwell time at the maximum load. The compliance of the machine was determined to be 0.648 μm/N and subtracted from the penetration data. Three ball indenters were used for the measurements: 400 μm in diameter hemispherical diamond tip, 1 mm diameter spherical tungsten carbide ball mounted indenter and a 2 mm diameter hardened steel ball mounted indenter.

Indentations were scanned using a laser non-contact profilometer (Zygo model New View with Metropro software). After defining the reference 'ground', the area scan readily yielded the maximum height and volume of the mound (pile-up), and the depth and the volume of the pond (crater).

3. Results

Room temperature fracture toughness K_c versus the average coordination number $\langle r \rangle$ from Vickers-produced indentations is shown in Fig. 1 for the binary and ternary systems. A pronounced maximum is observed in K_c value at $\langle r \rangle = 2.4$ for the binary system ($GeSe_4$), but not in the

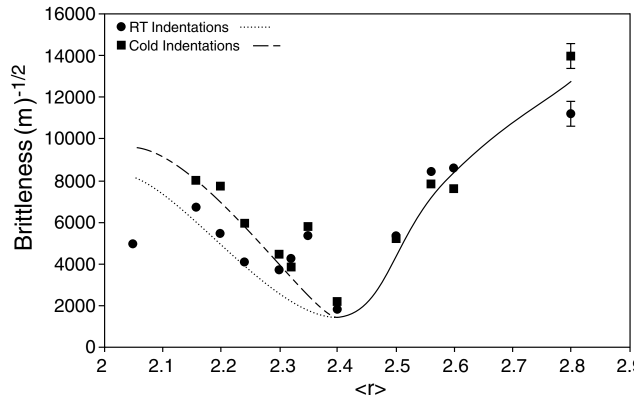

Fig. 2. Brittleness versus $\langle r \rangle$ for binary Ge–Se at room temperature (RT) as well as at −80 °C.

ternary. Calculated brittleness B versus $\langle r \rangle$ for the binary system is shown in Fig. 2. Brittleness for the binary system is high for the low $\langle r \rangle$ glasses and decreases to a minimum at $\langle r \rangle = 2.4$ ($GeSe_4$ composition), before increasing again. Toughness maximum at $\langle r \rangle = 2.4$ has also been observed by Guin et al. [21], however, they report the lowest B for 15–20%Ge glasses. The minimum in B is observed in both room temperature and cold indentation (−80 °C) data. Cold indentation hardness, hence brittleness, for glasses away from the $\langle r \rangle = 2.4$ composition appears to be higher than that at room temperature, particularly so for the low $\langle r \rangle$ glasses. No such distinct minimum in brittleness was observed for ternary glasses (not shown).

A typical plot of load P versus penetration h and an idealized view of a Brinell indent at the point of maximum load and after recovery are shown in Figs. 3(a) and 3(b). Fractional elastic recovery is greater in the depth direction than in the surface plane, as a result, the contact surface radius is larger than that of the indenter. Variation of the Brinell hardness-under-load H_{BL} with $\langle r \rangle$ for the Ge–Se binary using 1.5 kg load is shown in Fig. 4. Also shown is the comparison between the 0.4 mm and 2 mm diameter Brinell indenters. (Since the data for the 1 mm diameter indenter were invariably intermediate to those for the

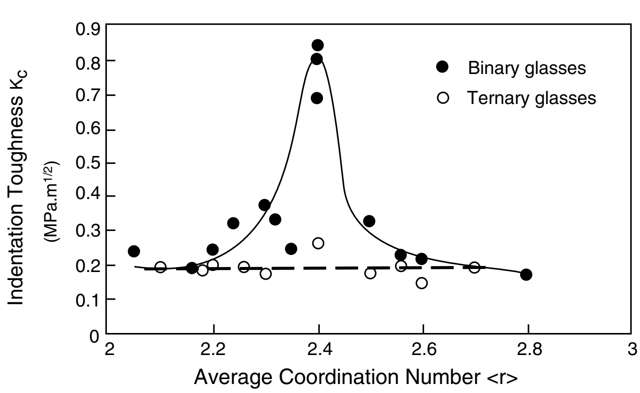

Fig. 1. Fracture toughness, measured at room temperature, versus $\langle r \rangle$ for both binary and ternary systems. Curves are drawn as a guide to the eye.

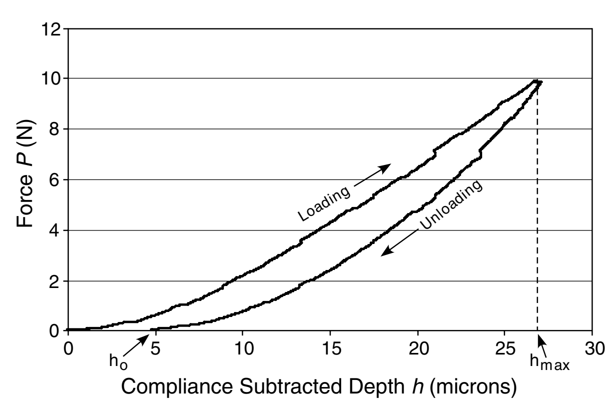

Fig. 3(a). Plot of load P versus penetration h (instrument compliance corrected) during loading as well as unloading.

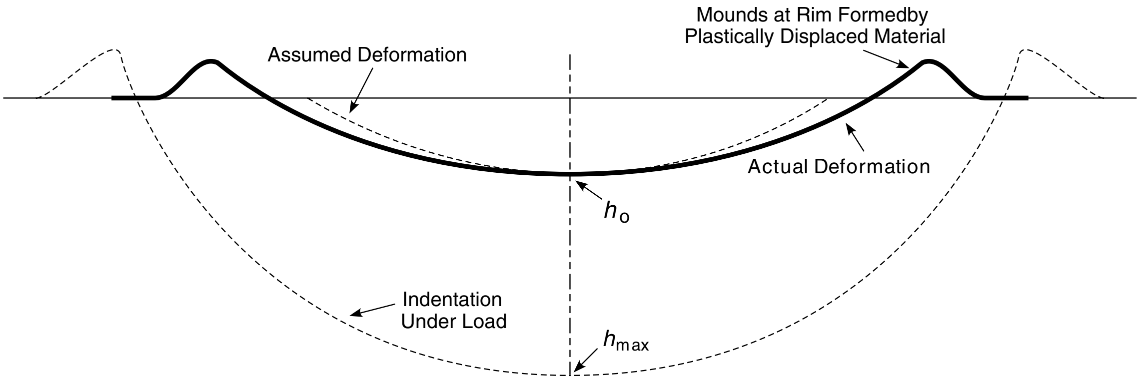

Fig. 3(b). Schematics of a Brinell indentation. Some material is sheared during indenting and forms a mound around the rim. Upon unloading, the elastic recovery of the depth dimension is larger than that of the surface plane resulting in the ‘pond’ becoming shallower than the indenter geometry.

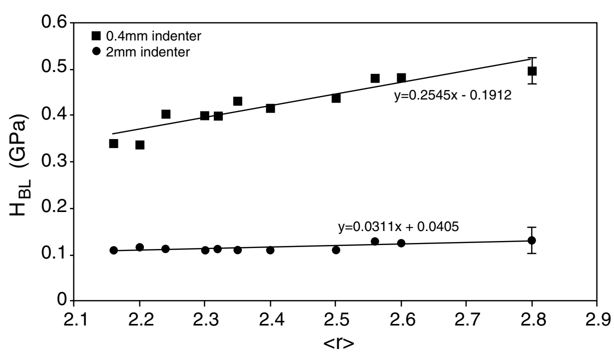

Fig. 4. H_{BL} versus $\langle r\rangle$ for the binary Ge–Se system shown for two indenters: $D = 400$ μm and 2 mm; and maximum load = 1.5 kg. Lines are least-squares fits.

0.4 mm and the 2 mm indenters, these are not plotted for clarity.) Least-squares fitted H_{BL} for the ternary system could be described by H_{BL} (GPa) $= 0.241\langle r\rangle - 0.175$ for the 0.4 mm indenter and H_{BL} (GPa) $= 0.158\langle r\rangle - 0.162$ for the 2 mm indenter. Values of H_{BL} obtained with a smaller diameter indenter are higher than those obtained with a larger diameter indenter; apparently D dominates in the calculation of H_{BL}. Figs. 5(a) and (b) show the variation of Brinell hardness-from-unloaded-depth H_{BD} with $\langle r\rangle$ for the binary system at two different loads. Because the ratio of the unloaded depth to the projected impression diagonal is smaller than that based on the indentation geometry as seen in Fig. 3, the H_{BD} values are greater than those measured using static microhardness machines (data reported in Ref. [5]) and much greater than H_{BL}. H_{BD} increases with $\langle r\rangle$ for both binary and the ternary (not shown). Whereas, in the binary system, there is a local maximum observed in H_{BD} at the physical threshold $\langle r\rangle = 2.4$ for both indenters (400 μm and 2 mm), the H_{BD} in ternary system appears to increase monotonically (not shown).

Figs. 6(a), (c), and (e) show a laser area scan of the indentations on $\langle r\rangle = 2.2$, 2.4, and 2.56 binary Ge–Se glasses using 400 μm indenter and 1 kg load.[2] The corresponding traces across the indent are shown in Fig. 6(b), (d), and (f) (caution: the width scale is much coarser than the height scale). One may observe that both the low $\langle r\rangle$ and the high $\langle r\rangle$ glasses show cracking. In the low $\langle r\rangle$ glasses, cracking appears to be mostly ringlike around the

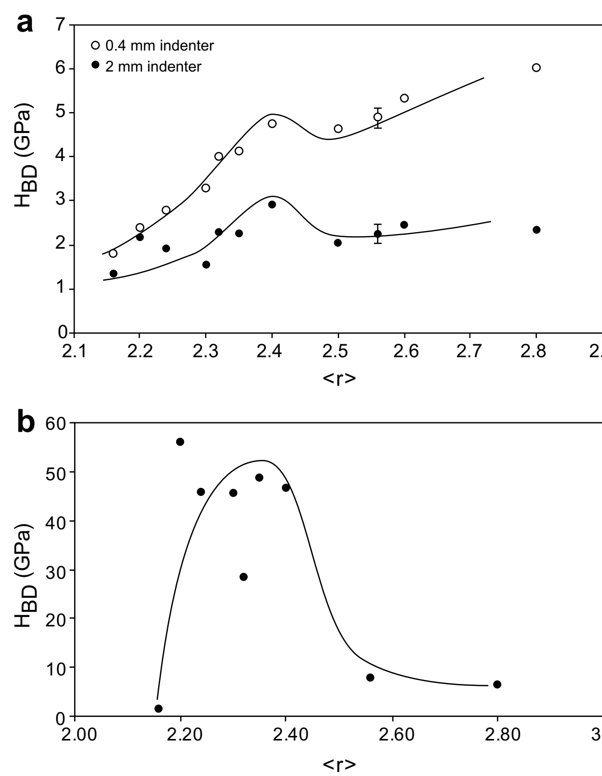

Fig. 5. H_{BD} versus $\langle r\rangle$ for the binary Ge–Se system shown for two indenters: (a) $D = 400$ μm and 2 mm; and maximum load = 1.5 kg; and (b) $D = 400$ μm; and maximum load = 1 kg. Curves are drawn as a guide to the eye.

[2] The reader is encouraged to check out topological similarities of these indentations with that of the Barringer crater in Winslow, AZ (see: www.barringercrater.com).

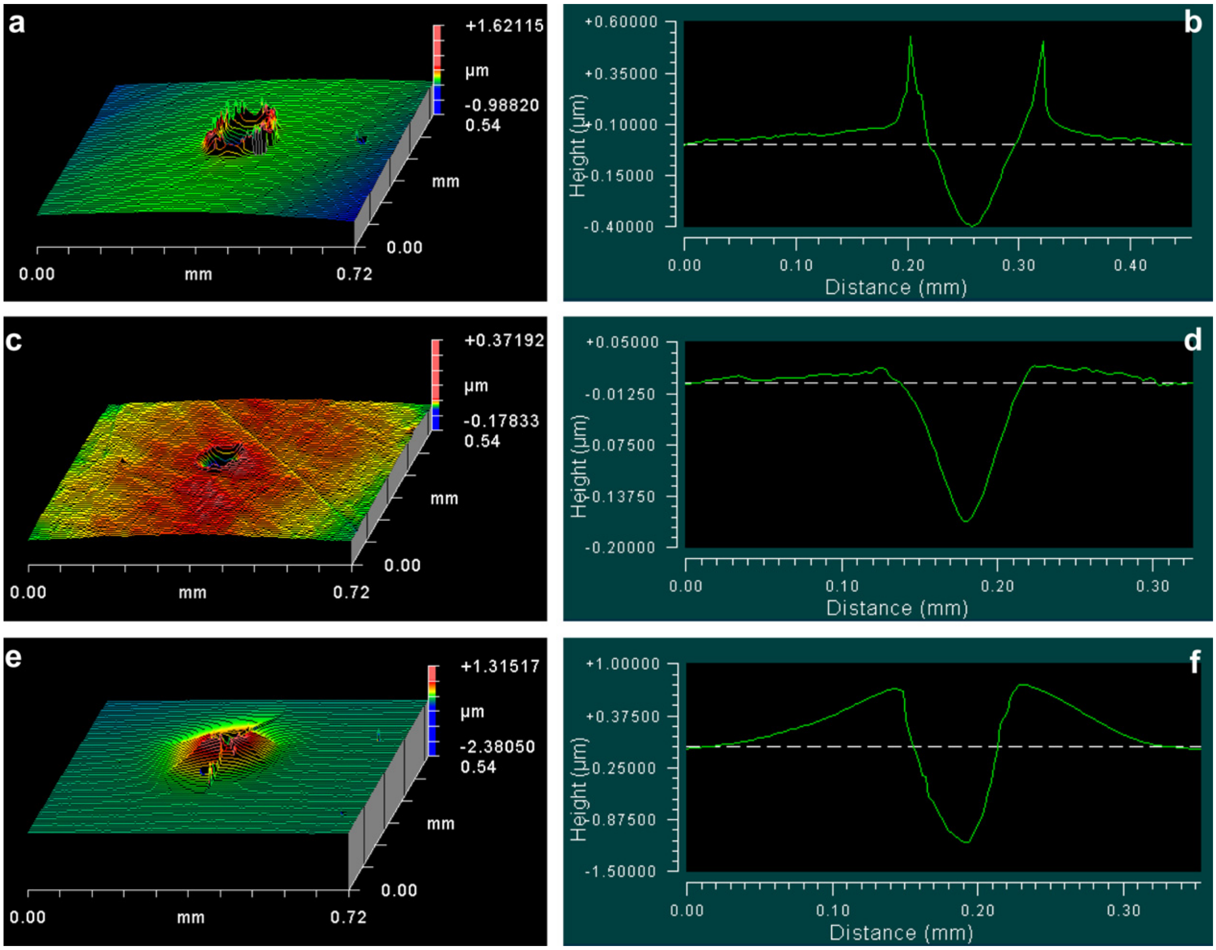

Fig. 6. (a)–(f) (left to right): Laser profilometer scans (a, c, and e) and corresponding traces (b, d, and f) of indentation after indenting the binary glasses with 400 µm diameter Brinell indenter at a maximum load of 1 kg. (a and b) $\langle r \rangle = 2.20$; (c and d) $\langle r \rangle = 2.40$; and (e and f) $\langle r \rangle = 2.56$.

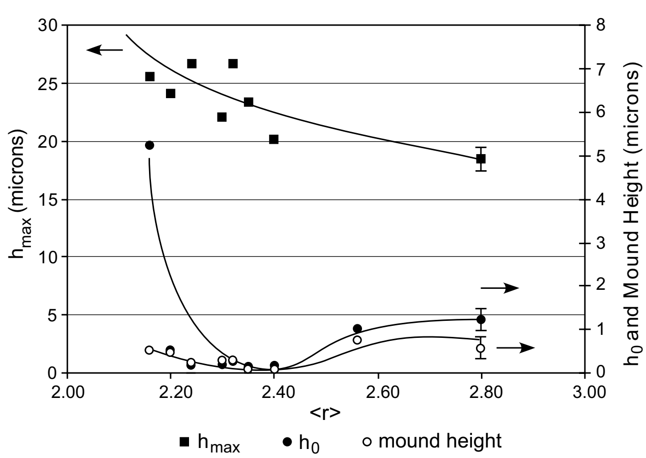

Fig. 7. Plots of h_{max}, h_o and the mound height as a function of $\langle r \rangle$. Curves are drawn as a guide to the eye.

mound. On the other hand, mostly radial cracks are observed in the high $\langle r \rangle$ glasses. Cracking is distinctly absent in the $\langle r \rangle = 2.4$ glass. Fig. 7 shows the h_{max}, h_o, and mound height against $\langle r \rangle$; values of the mound and the pond volumes after unloading, and the fractional unrecovered pond volume are shown plotted against $\langle r \rangle$ in Fig. 8(a)–(c).

4. Discussion

All of the measured indent parameters, namely, h_{max}, h_o, mound height, values of the mound and the pond volumes after unloading, and the fractional unrecovered pond volume show minima when plotted against $\langle r \rangle$ around $\langle r \rangle = 2.3$–2.4 for the binary system. Note also that the fractional unrecovered volume is only about ~8% for the lowest $\langle r \rangle$ glass and as little as 0.3% for $GeSe_4$, suggesting that the indenting process in these glasses is mostly elastic, at least, at lower loads. This is in contrast to the observations of Guin et al. [22] who reported only ~60% of the deformation to be elastic, presumably because they used a Vickers indenter whose sharp edges caused the material to exceed its yield strength. There is, yet, some measurable plasticity which is minimized at the $\langle r \rangle \cong 2.4$ binary. It is also clear that the mound volume is generally higher than the corresponding pond volume after unloading (Fig. 8(a) and (b)). One conclusion from this apparent disparity is that the mounds have lower density than the rest of the material:

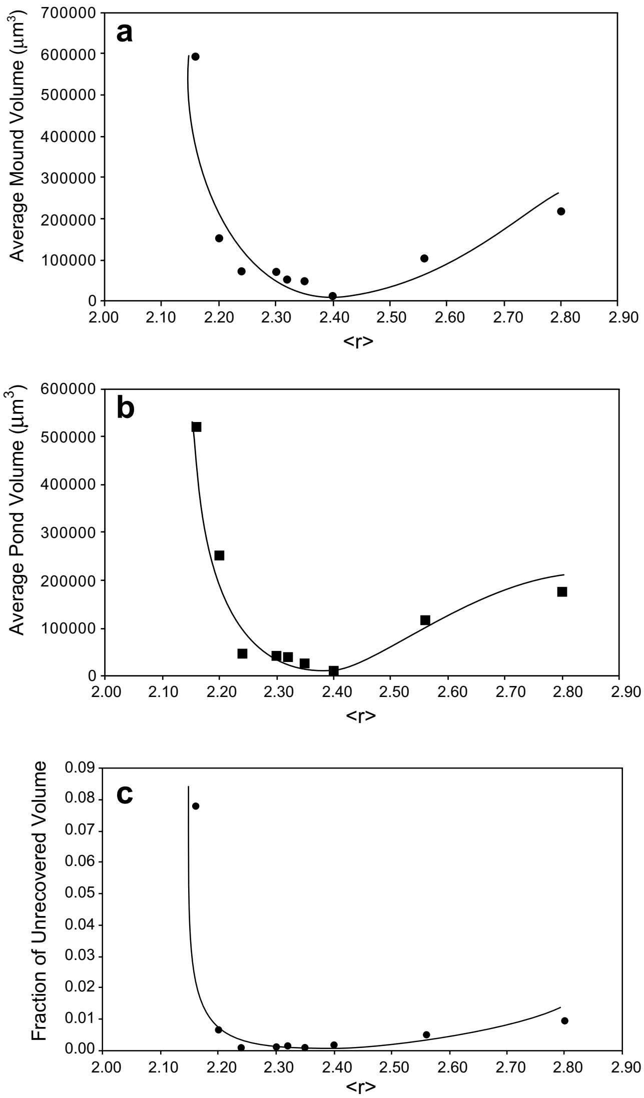

Fig. 8. (a) Mound volume, (b) pond volume, and (c) fraction unrecovered volume versus $\langle r \rangle$.

presumably, cracking allowed lifting of material 'plates' during the indentation such that some hollowness exists within the mounds. A similar analysis of the ternary glasses is not completed so far.

The maximum in H_{BD} (Fig. 5) is clearly due to a minimum in the final penetration depth (h_o). Because the lower load produces much smaller residual impression, hence, H_{BD} is much higher for smaller loads (Fig. 5(b)). It stands to reason that variation in the deformation behavior, hence of indentation cracking, in a given material arises likely from variation in experimental conditions of loading such as the indenter shape, size, maximum load and, presumably, the loading rate used. Clearly, H_{BD} using a spherical indenter would lose meaning if the load were small enough to have nearly complete recovery.

One may also argue that indenter-produced deformation in binary chalcogenide glasses is mostly shear plasticity-driven in glasses away from the $\langle r \rangle = 2.4$ composition, while in the $\langle r \rangle = 2.4$ composition ($GeSe_4$ glass), the high connectivity transforms the applied load to larger fraction of hydrostatic stress components. The fraction of the hydrostatic stress which is beyond the hydrostatic yield limit produces densification such that very little material is pushed out in shear mode to form a mound around the indent in $GeSe_4$.

The apparent lack of extrema in the ternary chalcogenide system is puzzling. It is likely that the term 'average coordination number' itself is a simple representation of the connectivity of the network and does not give a complete picture of the network structure. This is evident when the two glasses $Ge_{20}Se_{80}$ ($\langle r \rangle = 2.4$) and $Ge_{12.5}Sb_{15}Se_{72.5}$ ($\langle r \rangle = 2.4$) are compared: the former shows 'magic trends' whereas the latter does not. In order to have a better understanding of the network structure, particularly in relation to binary versus ternary (or multicomponent) compositions, the bonding between individual atoms need to be studied. *Ab initio* simulations of glass structure should, perhaps, help.

5. Conclusions

The conclusions drawn from this work can be summarized as follows:

- Within the Ge–Se binary glass family, the $GeSe_4$ glass ($\langle r \rangle = 2.4$) shows minima in indentation depth, mound height around the rim, radial cracking from the edge of the rim, and brittleness; and maxima in hardness-from-unloaded-depth (H_{BD}), and indentation toughness.
- During indentation, a given material deforms and cracks differently depending upon the indentation shape, size, maximum load, and the loading rate employed.
- The origin of the maximized H_{BD} is the maximization of fractional elastic recovery, or the yield strength, at $\langle r \rangle = 2.4$. This is consistent with the arguments of optimized connectivity of the $GeSe_4$ glass.
- The reduced radial cracking, hence the increased indentation toughness, in turn, give rise to decreased brittleness of the $GeSe_4$ glass.
- Ternary Ge–Sb–Se glasses do not show any such trends.

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